



南京大學
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PALM: Synergizing Program Analysis and LLMs to Enhance Rust Unit Test Coverage

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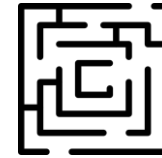
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Traditional Methods (SBST, Concolic)

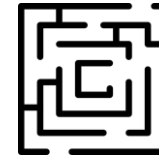


- Struggle with Rust's complexity (ownership, concurrency)
- Suffer from **Path Explosion** and **Environment Modeling** issues
- **Result:** Limited coverage for complex logic

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Existing LLM Methods

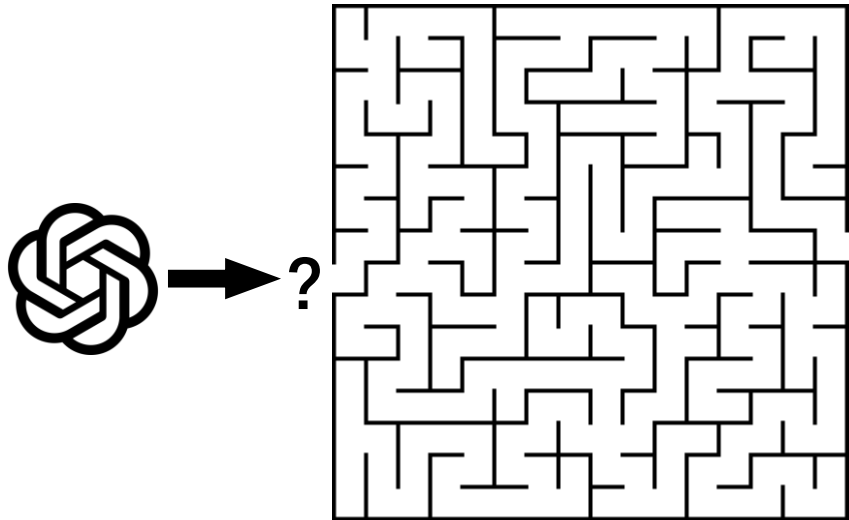


- Rely on **fixed, generic prompts**
- Lack specific guidance on internal logic
- **Result:** Low compilation rates & poor branch coverage

Our Insight: Guide the LLM with a Path-Specific Map

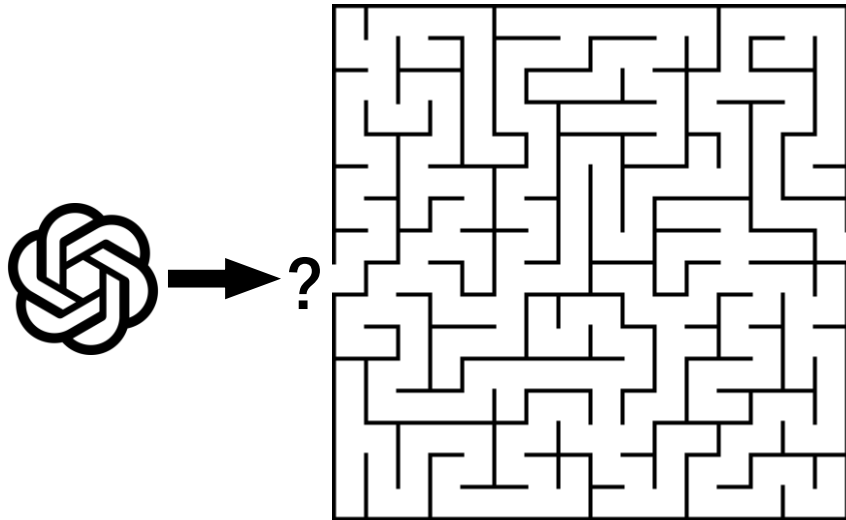


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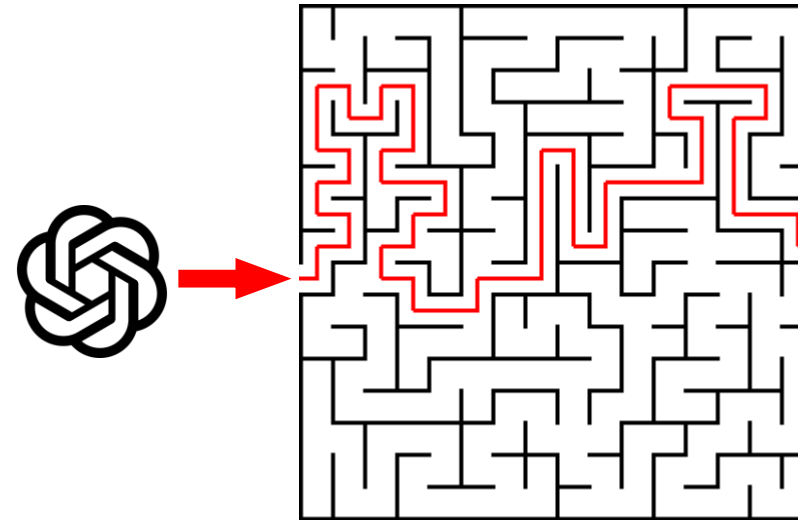


Common Approach: Ask the LLM to generate a whole test suite at once.

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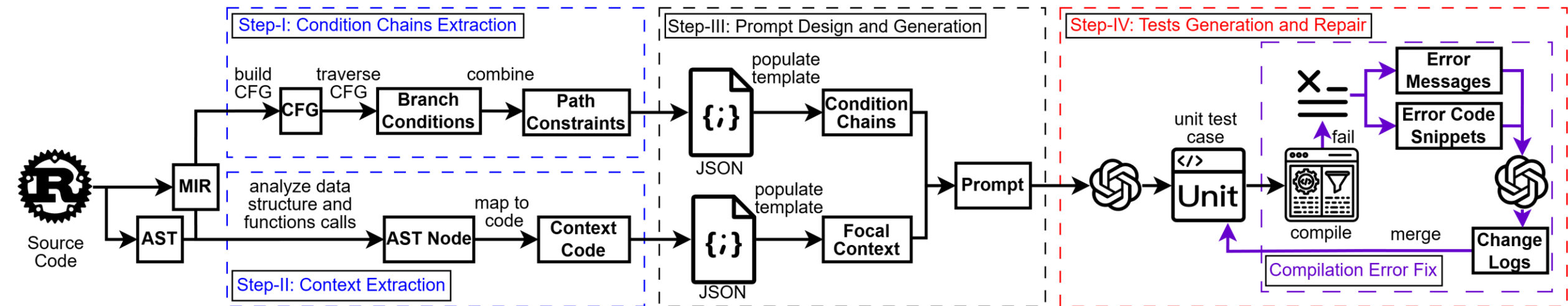


Our Approach: Guide the LLM with a specific mission for each execution path.

PALM: A Path-Guided, LLM-Powered Test Generator



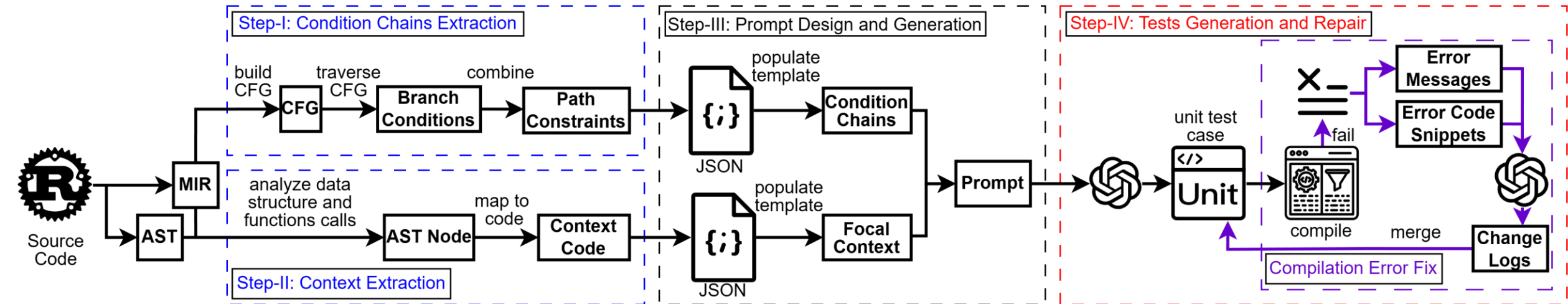
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1. Path & Constraint Extraction

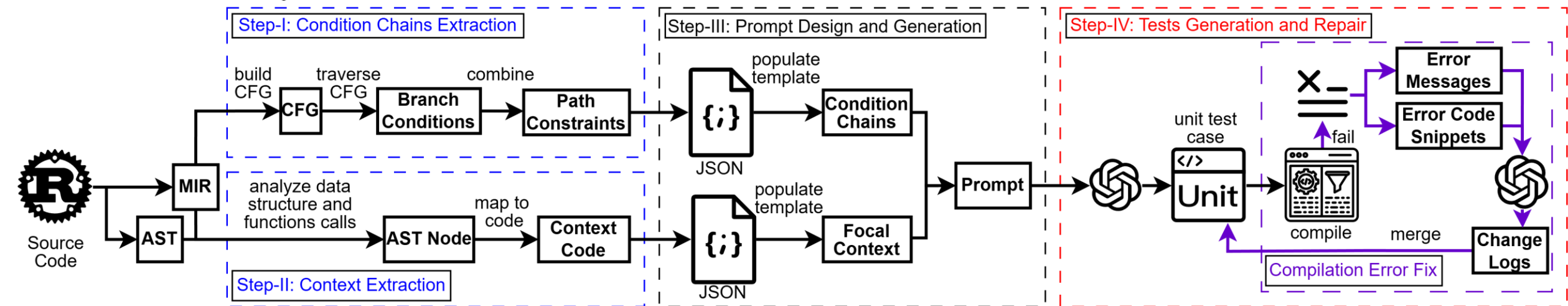
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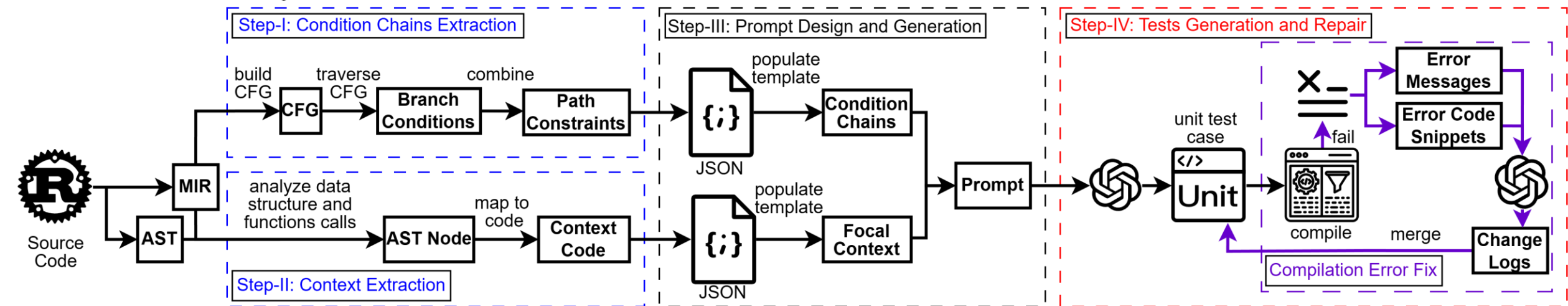
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Create a specific prompt for each path.

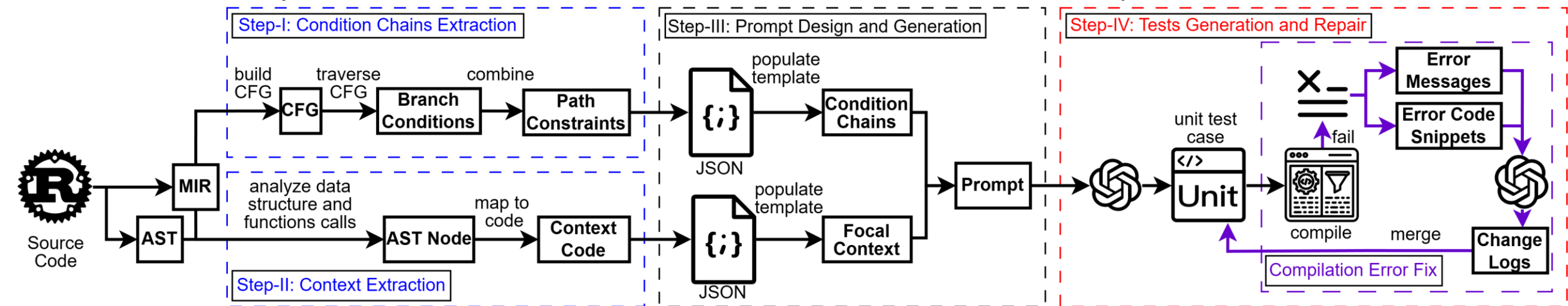
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1. Path & Constraint Extraction

Analyze MIR/HIR to get precise condition chains.

4. Iterative Repair Loop

Use compiler feedback to fix errors automatically.



2. Context Extraction

Gather relevant struct/trait definitions.

3. Path-Guided Prompting

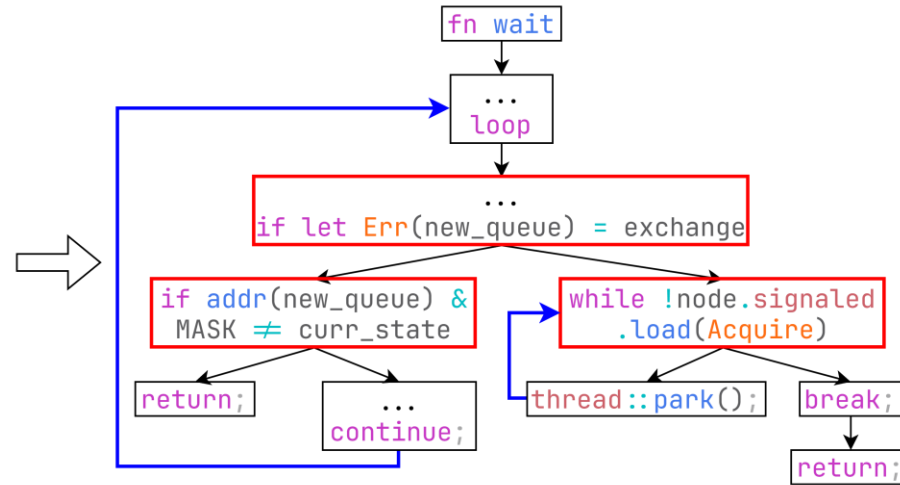
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Program Analysis using MIR/HIR



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```
fn wait(queue, mut curr_queue) {
  ...
  loop {
    ...
    if let Err(new_queue) = exchange {
      if addr(new_queue) &
        MASK != curr_state {
        return;
      }
      ...
      continue;
    }
    while !node.signaled.load(Acquire) {
      thread::park();
    }
    break;
  }
}
```

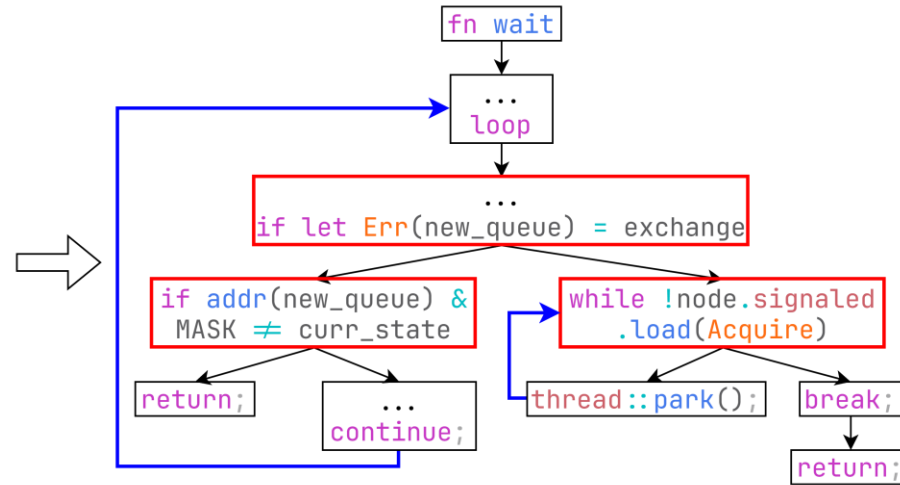


Condition Chains Extraction

- Analyze **MIR** for precise CFG →
- Traverse CFG to extract paths →
- Map back to **HIR** to recover source-level semantics →
- **Output:** Fine-grained **Condition Chains**

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<pre>use std::cell::{Cell, UnsafeCell}; ...</pre>	Uses
<pre>const INCOMPLETE: usize = 0x0; const RUNNING: usize = 0x1; ...</pre>	Consts
<pre>struct Waiter { thread: Cell<Option<Thread>>, signaled: AtomicBool, next: *mut Waiter, }</pre>	Structs
<pre>fn wait(queue, mut curr_queue) { ... }</pre>	Focal Method
<pre>mod strict { pub(crate) fn addr<T>(ptr: *mut T) → usize where T: Sized, { unsafe { core::mem::transmute(ptr) } } }</pre>	Other Methods

Context Extraction

- Identify all dependencies of the focal method:
 - Struct & Enum definitions
 - Trait implementations
 - Called functions
- Provide necessary context to ensure valid code generation.

Guided Generation and Automated Repair



Guided Generation and Automated Repair

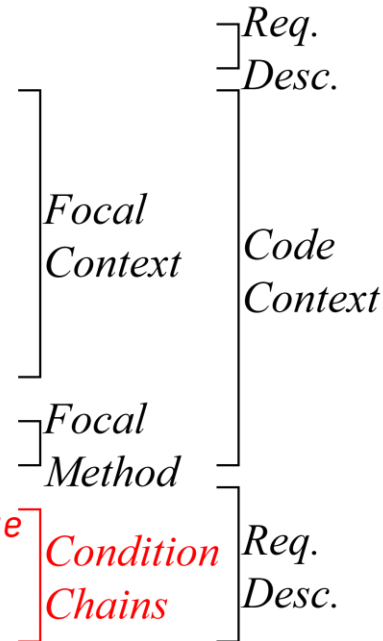
```
As a software testing expert...
use std::{...};
struct Waiter {...}
fn wait(...) {...}
mod strict {
    pub(crate) fn addr<T>(...) → usize
    ...
}
```

The function to be tested is presented:

```
fn wait(...) {...}
```

Given the following constraints...:

```
// constraint: exchange matches Err(...) is true
// constraint: addr(...) & MASK ≠ ... is false
// constraint: ...
```



Path-Guided Prompt

Prompt =

- Task Instruction (“As a software testing expert...”)
- Focal Method + Context
- Target Condition Chain:
 - // constraint: exchange matches Err(...) is true
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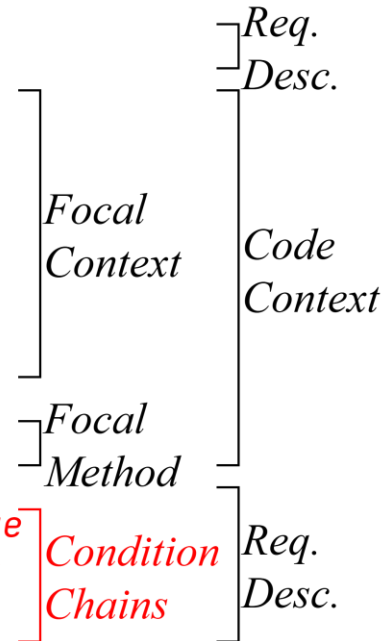
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 - // constraint: exchange matches Err(...) is true
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Error

```
#[test]
fn test_03() {
    ...
    wait(&queue, curr_queue);
    assert_eq!(queue.load(Ordering::Acquire)
                & STATE_MASK, COMPLETE);
}
```

Correct

```
#[test]
fn test_03() {
    ...
    wait(&queue, curr_queue);
    assert_eq!(queue.load(Ordering::Acquire)
                as usize & STATE_MASK, COMPLETE);
}
```

Automated Repair

Loop: Compile → Capture Error → Prompt LLM for Fix → Apply Patch

Before:load(...) & STATE_MASK (Type Error)

After:load(...) as usize & STATE_MASK (Fixed!)

Experimental Setup



Research Questions

- **RQ1: Effectiveness?**
(vs. Baselines)
- **RQ2: Ablation Study?**
(Impact of each component)
- **RQ3: Practical Usability?**
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Baselines

- **Classic:** RustyUnit (SBST),
RuMono (Fuzzing)
- **LLM-based:** RUG,
SymPrompt, ChatTester

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Subjects

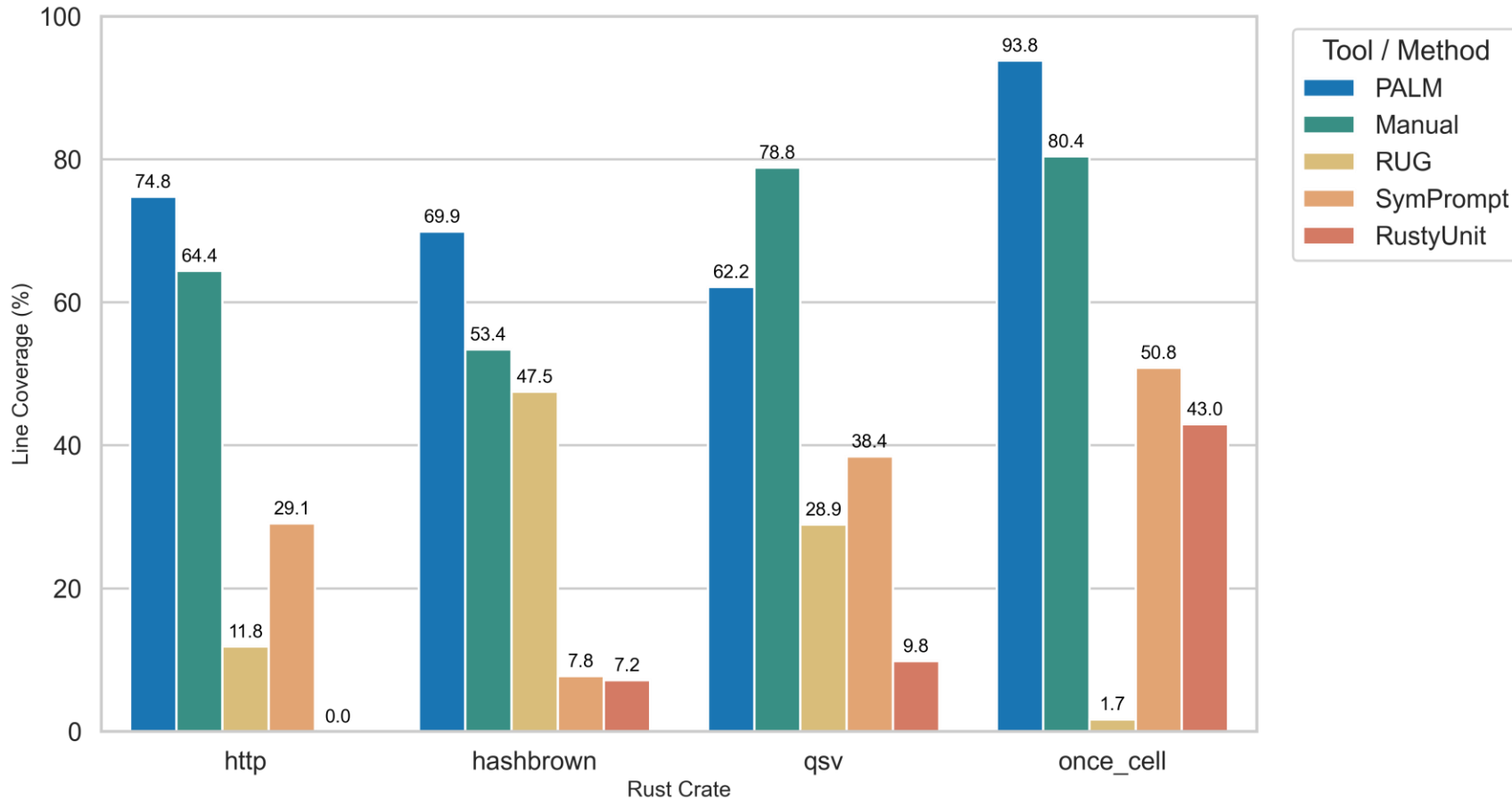
- **15** popular, mature Rust crates (e.g., serde, rand, regex, http, hashbrown)
- **4145** focal methods in total

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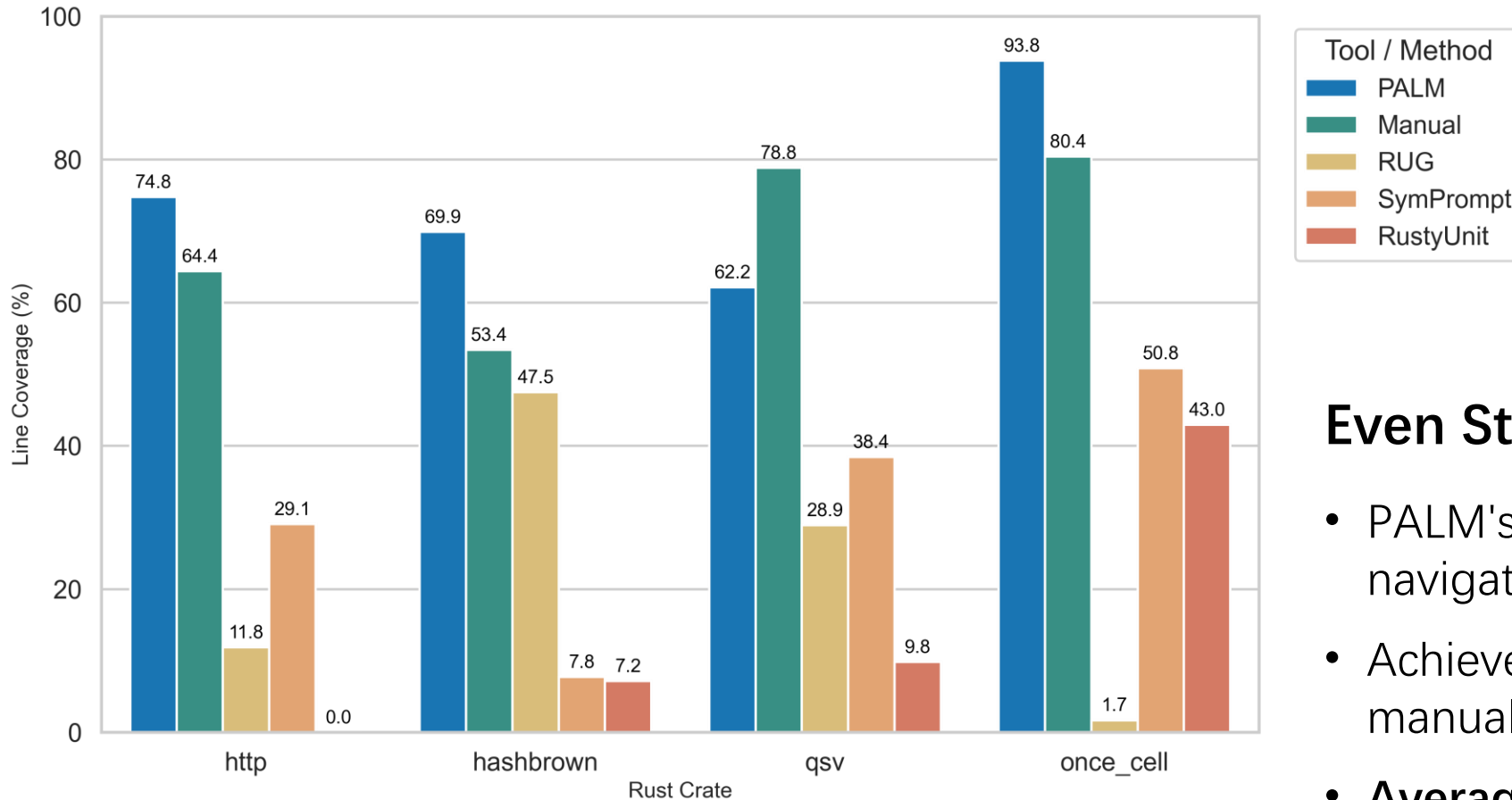
Line Coverage Comparison on Representative Crates



PALM's average line coverage (**72.30%**) is comparable to human effort (**70.94%**) and significantly outperforms other automated tools.

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Even Stronger on Branch Coverage

- PALM's path-guided approach excels at navigating complex conditional logic.
- Achieved **higher branch coverage** than manual tests in **10 out of 15** projects.
- **Average Branch Coverage:**
 - **PALM: 69.5%**
 - Manual: 66.5%

It Works in Practice! High Acceptance Rate in Open-source Projects.





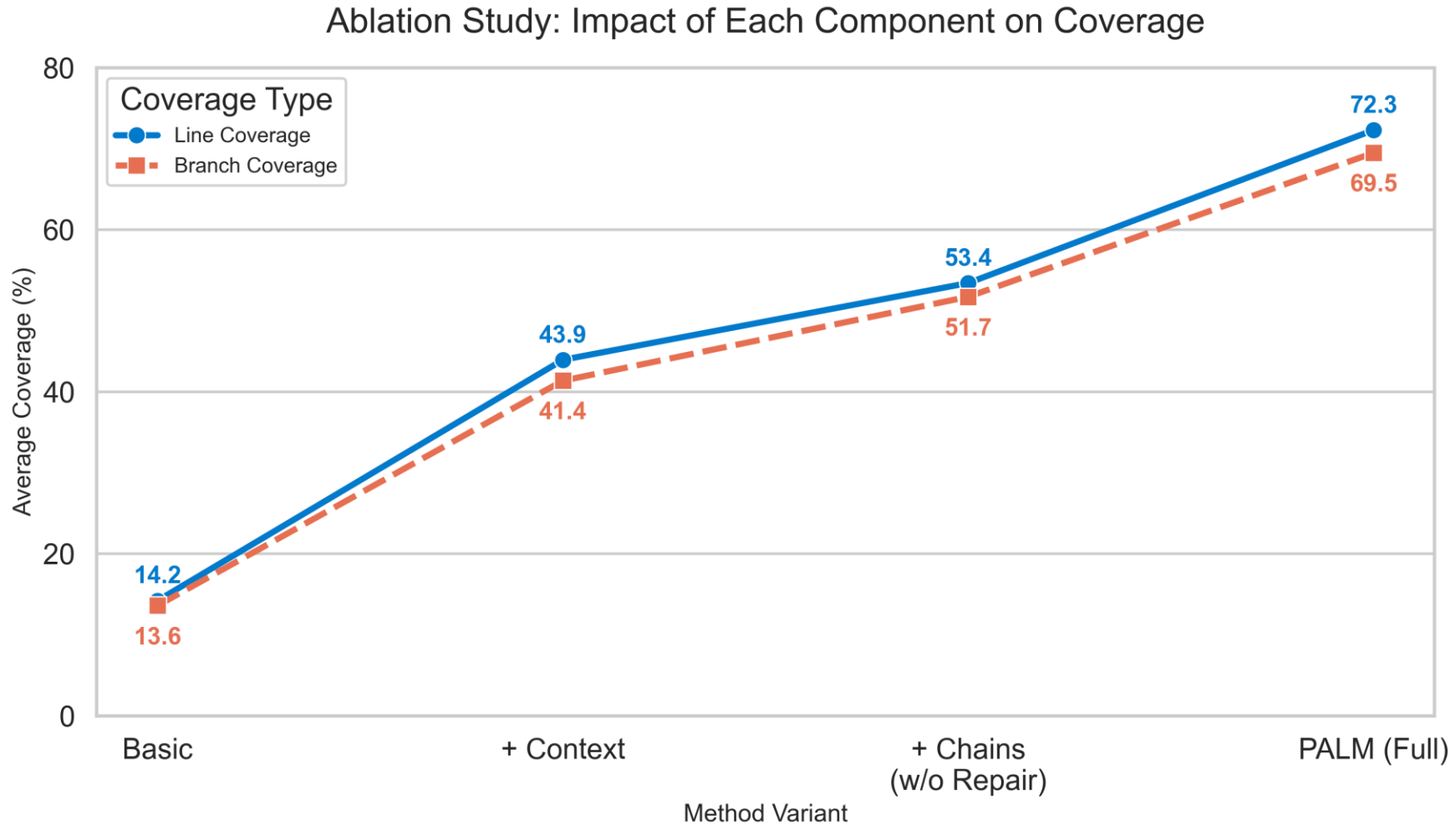
94.1%
Acceptance Rate

- Submitted **91** PALM-generated unit tests to 8 popular open-source projects.
- **80 tests were Accepted & Merged** by maintainers.
- Contributions to: hashbrown, once_cell, bytes, and more.

Ablation Study: Path-Guidance and Repair are Crucial



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Key Finding: Both **Path-Specific Guidance (Condition Chains)** and **Automated Repair** are crucial for achieving high performance.

Conclusion



- **PALM:** A novel approach synergizing program analysis and LLMs for Rust test generation.
- **Path-Guided Prompting:** Decompose the complex task into manageable steps, significantly improving coverage.
- **Iterative Repair:** Effectively fix compilation errors, making LLM-generated code practical and usable.
- **Proven Practicality:** Achieved coverage comparable to human effort and was validated by a **94.1%** pull request acceptance rate.

Thank You! Q&A

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